Autonomous Retail Research Agent

(An AI-Powered Multi-Agent Market Intelligence Platform)

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: AAI-01

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Introduction -

The rapid advancement of artificial intelligence has transformed how businesses collect and utilize data for decision-making, especially in the retail sector where market trends evolve quickly. Traditional research methods—based on manual data collection, static reports, and fragmented sources—are often slow, inefficient, and prone to gaps in insight. As a result, there is a growing need for intelligent systems that can automate the process of gathering, analyzing, and presenting market intelligence in a structured and reliable manner.

To address this need, the **Autonomous Retail Research Agent** is developed as an AI-powered multi-agent platform that automates end-to-end retail market research. The system combines live web data, a curated internal knowledge base, and vector-based memory to generate context-aware, professional research reports from a simple user query. By leveraging technologies such as CrewAI, LangChain, ChromaDB, and large language models, and delivering the solution through an interactive Streamlit interface, the platform enables faster, scalable, and more informed decision-making in the retail domain.

Problem Statement & Objectives

1. Problem Statement :

   Retail businesses today are overwhelmed by the volume and velocity of market data. Analysts and decision-makers struggle to gather timely, credible, and commercially relevant research on emerging trends, category opportunities, and margin dynamics. Traditional research methods rely on manual web browsing, static reports, and disjointed data sources—a process that is both slow and prone to information gaps.

   There is a clear need for an intelligent, automated system that can autonomously search live market data, cross-reference it against internal business heuristics, and produce structured, stakeholder-ready research reports—without requiring constant human intervention at every step.

2. Project Objectives :

   - To design and implement a multi-agent AI system capable of autonomously conducting live retail market research using real-time web data.
   - To integrate a curated internal knowledge base of retail frameworks (margin thresholds, 2026 trend keywords, trusted sources) that agents use as evaluation benchmarks.
   - To build a vector database memory layer using ChromaDB so that insights from previous research runs are retrievable and reusable in future sessions.
   - To generate structured, professional-grade market research reports in Markdown format, complete with comparison tables, executive summaries, and source attribution.
   - To enable export of the generated report in both .txt and .pdf formats with embedded charts and visual summaries.
   - To deliver all functionality through an interactive Streamlit web application, making the tool accessible to non-technical users.

3. Scope of the Project :

This project focuses on the retail and consumer goods domain. The system accepts a natural language query from a user and generates a comprehensive market research report. The scope includes:

- Live web research using Serper API for real-time market signals.
- Internal knowledge ingestion from three curated retail strategy documents covering margin frameworks, trend signals, and technology strategy.
- Vector-based retrieval from past reports stored in ChromaDB to enable contextual memory across sessions.
- Report generation with structured sections, a mandatory opportunity comparison table, and data visualisations (margin chart and trend alignment chart).
- PDF and text export functionality for report sharing and submission.

The system is designed for academic demonstration, high-level business design discussions, and collaborative report preparation. Deployment on Streamlit Cloud is planned as the final delivery stage.

Proposed Solution

1. Key features :

- Multi-Agent Research Pipeline: A CrewAI-powered sequential workflow with two specialised agents — Retail Researcher and Report Writer — that collaborate to produce a complete research report.
- Knowledge Base Integration: A structured KnowledgeStore built from internal .txt and .pdf files containing retail margin frameworks, 2026 trend keywords, and trusted domain references.
- Vector Memory Retrieval: ChromaDB-backed persistent vector store that embeds past reports and knowledge documents, allowing agents to retrieve semantically relevant context for each query.
- Live Web Research: Serper API integration gives the Retail Researcher agent access to real-time market data, news, and industry reports on the internet.
- Structured Report Generation: The Report Writer agent produces a consistent markdown report with sections including Executive Summary, Market Landscape, Key Opportunities, Opportunity Comparison table, Risks, Recommendations, and Sources.
- Data Visualisations: Automatically generated Matplotlib charts — Estimated Margin by Opportunity (bar chart) and Trend Alignment Score (line chart) — embedded in the report and available for download.
- PDF Export: ReportLab-powered PDF generation with embedded charts, summary image, comparison tables, and professional typography.

- Streamlit Web Interface: A polished, user-friendly application with query input, realtime agent activity logs in the sidebar, tabbed report display (Report / Visuals / Downloads), and download buttons.
- Report Persistence: All generated reports are saved with timestamped filenames in internal_repository/, enabling historical lookup and reload.

2. Overall Architecture / Workflow

The application follows a five-stage pipeline from user input to downloadable report:

Stage 1 – Knowledge Ingestion :-

At startup, LangChain DirectoryLoader reads all .txt and .pdf files from the knowledge_base/ directory. These are parsed into a KnowledgeStore object containing target profit margins, 2026 trend keywords, trusted research domains, and source quality rules.

Stage 2 – Vector Database Build :-

Knowledge base documents and any previously generated reports in internal_repository/ are chunked using LangChain's RecursiveCharacterTextSplitter and embedded using Google Gemini Embedding or OpenAI Embeddings. These embeddings are persisted to a ChromaDB vector store on disk.

Stage 3 – Query Input & Retrieval :-

The user enters a retail research query through the Streamlit interface. Before agents are invoked, a semantic similarity search retrieves the top-6 most relevant chunks from the vector store, forming a RetrievalMemory context that is passed to the agents.

Stage 4 – Multi-Agent Execution (CrewAI Sequential Pipeline) :-

The Retail Researcher agent performs live web searches via Serper, cross-references findings against the KnowledgeStore benchmarks and retrieved vector memory, and produces a structured research brief. The Report Writer agent then converts this brief into a professional Markdown report following a mandatory section template.

Stage 5 – Asset Generation & Export :-

The StorageService saves the markdown report with a timestamp. The system also generates: a summary banner image, a margin bar chart, a trend alignment line chart, and a professionally formatted PDF. All files are accessible for download from the Streamlit Downloads tab.

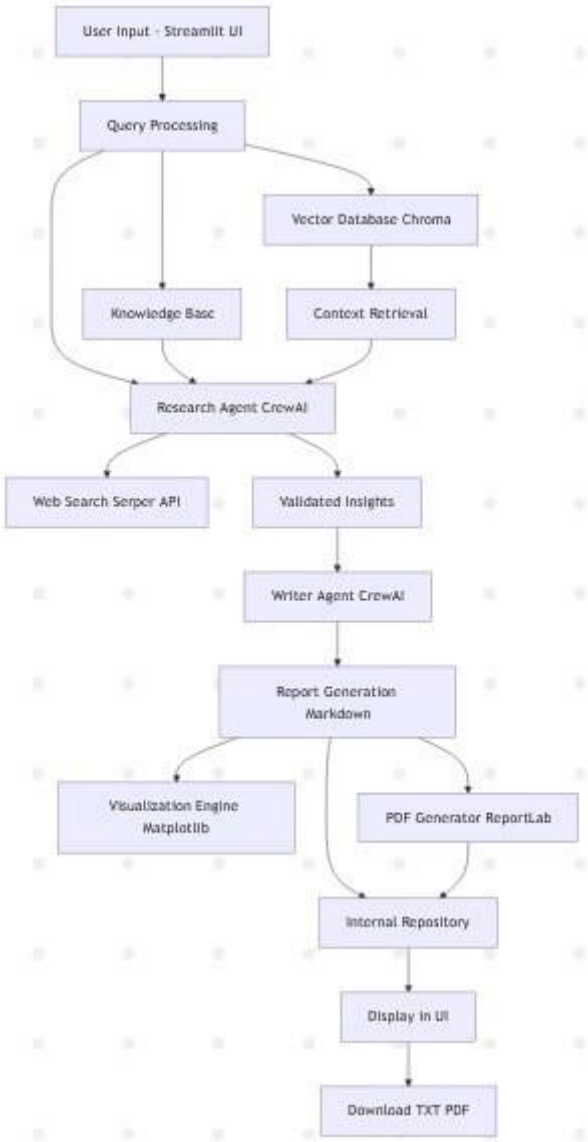


Fig 2.1 : Application Design

3. Tools & Technologies Used :

Component	Technology / Tool	Purpose
Frontend / UI	Streamlit	Web application interface for query input, report display, and downloads

Multi-Agent Orchestration	CrewAI	Sequential agent pipeline management (Researcher → Report Writer)
LLM Backbone	Google Gemini 2.5 Flash / GPT4o	Natural language reasoning for research and report generation
Knowledge Base Ingestion	LangChain (DirectoryLoader)	Loads .txt and .pdf knowledge files into structured context
Vector Database	ChromaDB + LangChain	Persistent embedding store for retrieval-augmented memory
Embeddings	Gemini Embedding / OpenAI Embeddings	Converts document chunks into vector representations
Web Research Tool	Serper API (via CrewAI Tools)	Live internet search for real-time retail market signals
Report Visualisation	Matplotlib	Bar and line charts for margin and trend alignment
PDF Generation	ReportLab	Export-ready PDF report generation
Environment Config	Python-dotenv	Secure API key management via .env file

Component Design - The component design is based on a modular architecture where different components handle specific tasks such as user interface (Streamlit), multi-agent processing (CrewAI), knowledge retrieval (LangChain), and vector storage (ChromaDB). Each component interacts through a defined pipeline, ensuring smooth data flow from user query to final report generation. This separation of concerns improves scalability, maintainability, and system efficiency.

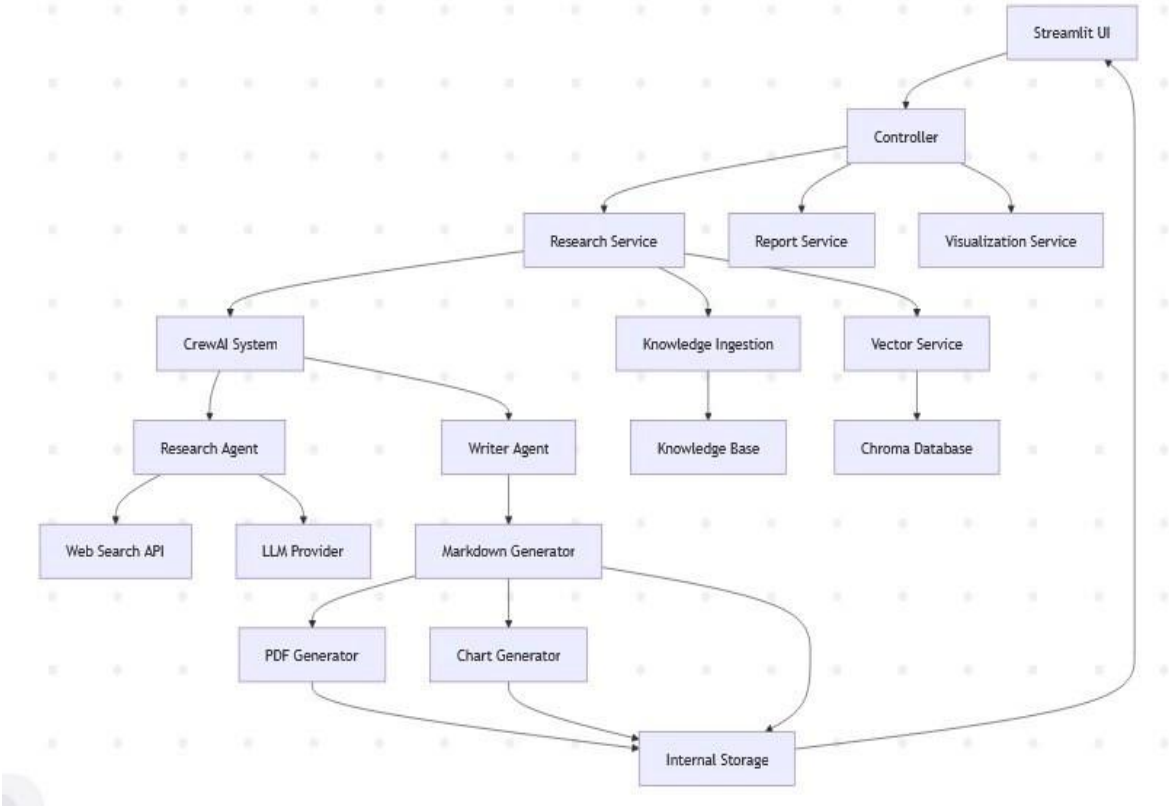


Fig 3.1 : Component Design

Database Design - The system uses ChromaDB as a vector database instead of a traditional relational database. It stores knowledge base content and past reports as embeddings, enabling semantic similarity search. This allows the system to retrieve contextually relevant information and support accurate, memory-driven report generation.

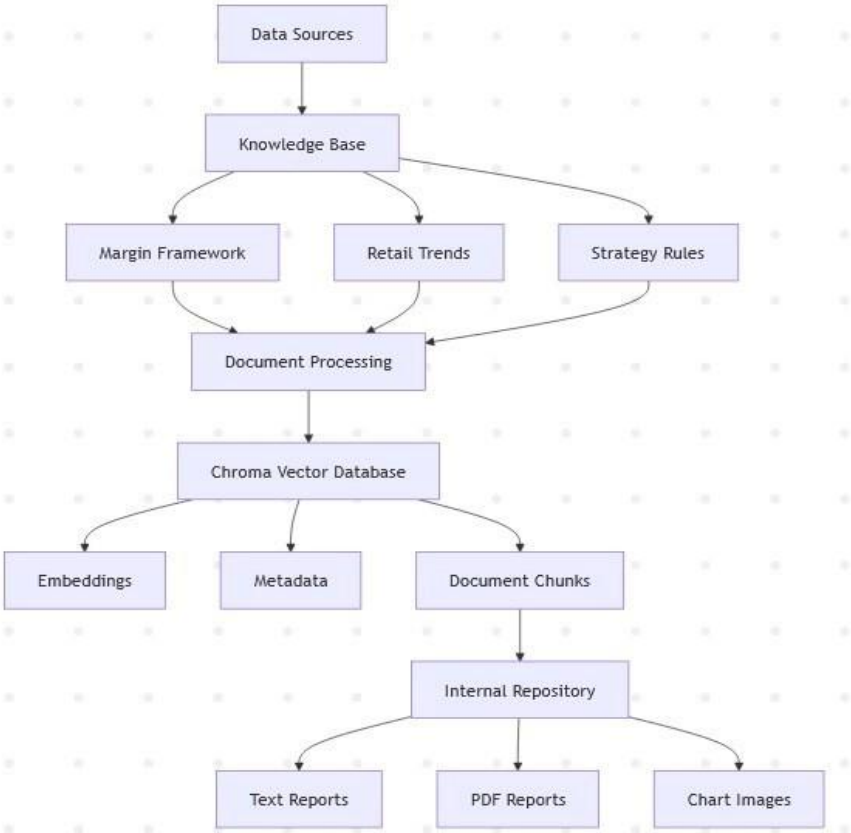


Fig 3.2 : Database Design

Class Diagram -

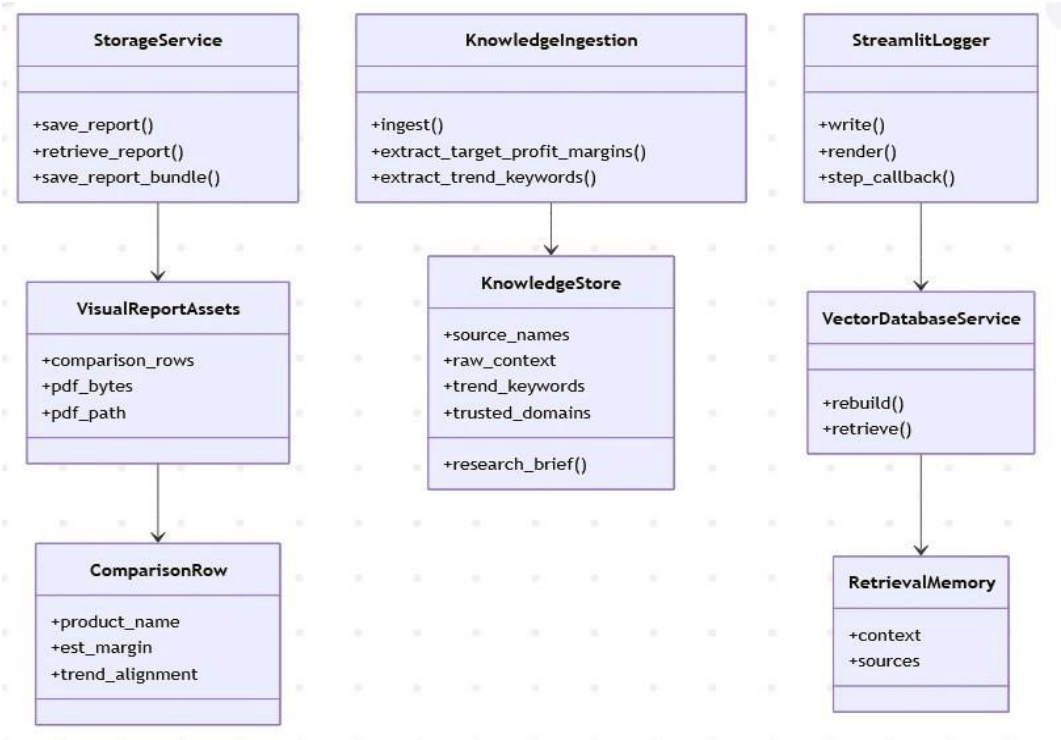


Fig 3.3 : Class Diagram

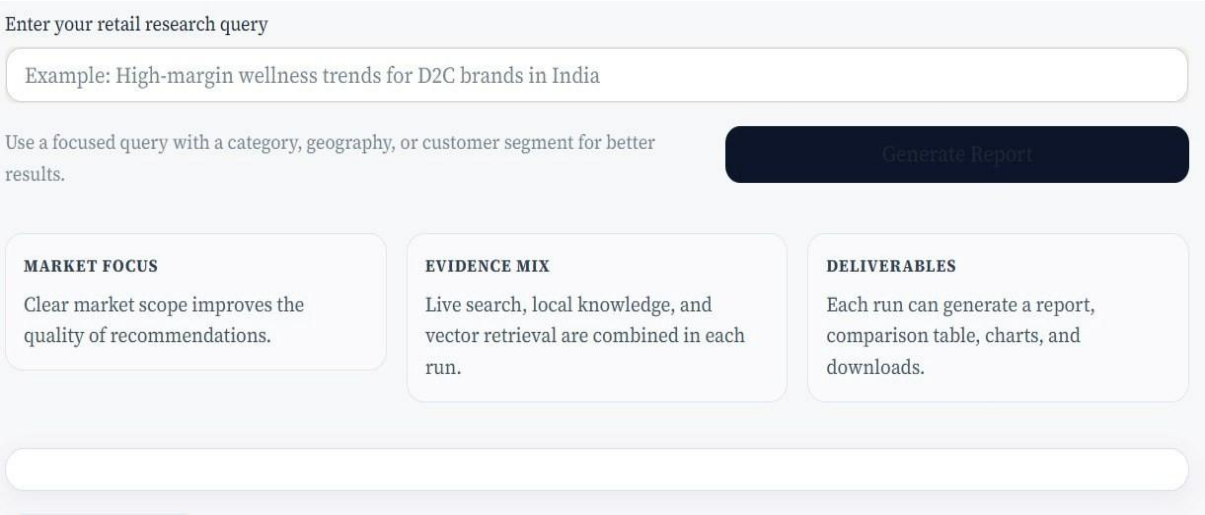
Results & Output

1. *Screenshots / outputs :*

The following screenshots demonstrate the working application deployed on Streamlit Cloud.

Screenshot 1: Application Hero Banner & Status Dashboard :

Screenshot 2: Research Workspace — Query Input :



Screenshot 3: Research Activity Log (Live Agent Output) :

Screenshot 4: Generated Report — Report Tab :

Market Research Report: Indian D2C Wellness Market Analysis 2026

Executive Summary

The Indian D2C wellness market is poised for significant growth in 2026, driven by high-margin trends such as digital-first personalization, subscription economy, and sustainable premiumization. Key opportunities include Wellness Nutrition, Mental Health Services, and Functional Fitness, with estimated margins ranging from 42% to 50%. This report analyzes the market landscape, identifies top growth categories, and provides strategic recommendations for stakeholders.

Market Landscape

The Indian D2C wellness market is expected to experience rapid growth in 2026, driven by increasing consumer demand for personalized and sustainable wellness products and services. The market is characterized by a shift towards digital-first personalization, subscription economy, and social commerce, with players leveraging short-form video commerce and AI-driven recommendations to drive engagement.

Screenshot 5: Visuals Tab — Margin and Trend Alignment Charts :

Opportunity Comparison

The following table compares the top opportunities in the Indian D2C wellness market:

Product Name	Est. Margin	Trend Alignment
Wellness Nutrition	45%	High
Mental Health Services	50%	Very High
Functional Fitness	42%	Medium

Screenshot 6: Downloads Tab — Export Options :

2. *Reports / dashboards / models :*

Each report generated by the system follows a fixed structured template and includes the following sections:

- Executive Summary — A concise 2-3 point summary of the most commercially significant findings.
- Market Landscape — Overview of the current state of the queried retail market or category.
- Key Opportunities — High-margin product and category opportunities identified from live research and knowledge base validation.
- Memory and Benchmark Signals — Retrieved insights from past reports and internal retail frameworks.
- Opportunity Comparison Table — A structured table with columns: Product Name | Est. Margin | Trend Alignment. Minimum 4 rows.
- Risks and Validation Notes — Source quality flags (Ground Truth vs. Weak Signal) and commercial risks.
- Recommendations — Actionable strategic suggestions for the queried market.
- Sources — All web and knowledge sources referenced during research.

In addition to the markdown report, the system automatically generates:

- A Summary Banner Image — A Matplotlib snapshot showing the query, executive summary points, and branding.
- An Estimated Margin Bar Chart — Comparing estimated gross margins across top identified opportunities.
- A Trend Alignment Line Chart — Scoring each opportunity (1–5) on alignment with 2026 retail trends.

3. *Key outcomes :*
    - The system successfully automates end-to-end retail market research: from a single text query to a downloadable, professionally formatted report.
    - The multi-agent architecture (CrewAI) enables a clear separation of concerns: the Researcher gathers and validates evidence, while the Writer ensures report quality and structure.
    - The vector memory layer (ChromaDB) allows the system to build compounding intelligence across sessions — earlier research informs future queries.
    - The knowledge base acts as a guardrail: agents are constrained to evaluate findings against proven retail margin thresholds and 2026 trend signals, reducing hallucination and improving commercial relevance.
    - The Streamlit interface makes the tool accessible to non-technical users with no coding knowledge required to generate a full research report.
    - The dual-format export (TXT and PDF) supports both collaborative workflows and formal academic or business submission requirements.

Conclusion :

The Autonomous Retail Research Agent successfully demonstrates how modern agentic AI frameworks can be applied to a real-world business intelligence problem. By combining CrewAI's multi-agent orchestration, LangChain's document ingestion and vector retrieval capabilities, the Serper API for live web research, and Streamlit for an accessible interface, the project delivers a working end-to-end research automation platform.

The core learning from this project is the practical understanding of how agents can be designed with specialised roles, backstories, and tools — and how they collaborate in a sequential pipeline to produce an output that neither agent could achieve alone. The integration of a persistent vector database introduces the concept of memory in AI systems, making each new research session more contextually grounded than the last.

The project also reinforced the importance of knowledge grounding: by anchoring the agents to an internal knowledge base of retail margin frameworks and trend heuristics, the system consistently produces commercially relevant outputs rather than generic summaries. This approach directly mirrors production-grade RAG (Retrieval-Augmented Generation) architectures used in enterprise AI applications.

Overall, this project provided hands-on experience with LLM API integration, prompt engineering for agentic tasks, vector database design, data visualisation pipelines, and professional report generation — all delivered within a deployable web application.

Future Scope & Enhancements :

While the current system is fully functional for retail research queries, several enhancements are planned or could be implemented to increase the platform's capability and commercial applicability:

1. Stronger Deterministic Report Templates

Introducing Pydantic output schemas or structured JSON contracts for agent outputs would make the report structure entirely predictable, removing any variability in section ordering or table formatting across different LLM providers.

2. Richer Visual Dashboards

Future versions could incorporate interactive Plotly or Alt air charts, allowing users to filter and explore opportunity data within the Streamlit interface rather than viewing static Matplotlib images.

3. Advanced Source Attribution & Confidence Scoring

Adding a confidence score to each research finding (based on source quality, domain credibility, and corroboration count) would help users quickly identify the most reliable insights in the report.

4. Expanded Knowledge Base with Sector Coverage

The knowledge base can be extended beyond retail to cover other sectors such as fintech, healthcare, or edtech — making the platform a general-purpose industry intelligence tool.

5. User Authentication & Multi-Tenant Support

With Streamlit authentication or a lightweight backend, the tool could support multiple users with separate report histories and private knowledge bases.

6. Integration with Internal Data Sources

Connecting the system to a company's internal sales data, CRM, or ERP through API connectors would enable personalised research that blends external market signals with proprietary business metrics.

7. Fine-Tuned LLM for Retail Domain

Fine-tuning a smaller open-source model on retail research corpora could reduce API costs and latency while improving domain-specific reasoning quality.